Variable-Length Records

Introduction

Variable-length records are commonly used in database systems due to:

- Storage of multiple record types in a file.
- Record types with variable-length fields (e.g., `varchar`).
- Record types with repeating fields (arrays, multisets in older models).

1 Representation of Variable-Length Records

Key Concept

Attributes are stored in order.

- Variable-length attributes are represented by a fixed-size pair `(offset, length)`.
- Actual attribute data is stored after all fixed-length attributes.
- Null values are indicated using a **null-value bitmap**.

1.1 Example Layout with Infographic

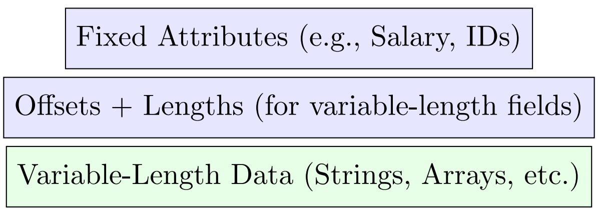

Figure 1: Logical Structure of a Variable-Length Record

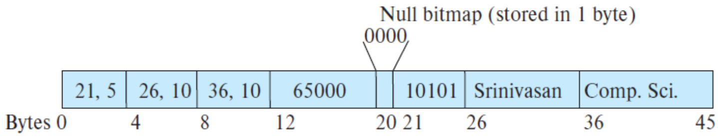

Figure 2: Variable-Length Record Example with Byte Offsets and Null Bitmap

Explanation of the Variable-Length Record Example

Record Explanation

- **Record Length:** Byte 0 to Byte 45 (46 bytes total).
- **Fixed Attributes:** Stored at the beginning for direct access. Example:
  - `21,5`: Offset = 21, Length = 5 bytes (ID)
  - `26,10`: Offset = 26, Length = 10 bytes (Name)
  - `36,10`: Offset = 36, Length = 10 bytes (Department)
  - `65000`: Salary stored in 8 bytes
  - `10101`: Another numeric identifier
- **Textual Data:**
  - "Srinivasan" (Instructor Name)
  - "Comp. Sci." (Department Name)
- **Null Bitmap:** 1 byte, between Byte 20–21. Value "0000" indicates first four attributes are not null.
- **Offsets + Lengths Section:** Metadata for variable-length attributes.
  - *Offset:* Starting byte of the attribute
  - *Length:* Size in bytes of the attribute
- **Variable-Length Data:** Stored consecutively after fixed-length attributes. Supports fast access, insertion, and deletion without moving the entire record.
- **Design Advantages:** Fast attribute extraction, minimal fragmentation, null attributes consume no storage.

2 Slotted Page Structure

Variable-length records are commonly stored using a **slotted page structure**.

2.1 Slotted Page Header Contains:

- Number of record entries.

- End of free space in the block.
- Location and size of each record.

2.2 Characteristics

- Records can be moved to keep them contiguous.
- Header entries are updated whenever records move.
- Pointers reference the header entry, not the record itself.

2.3 Diagram of Slotted Page

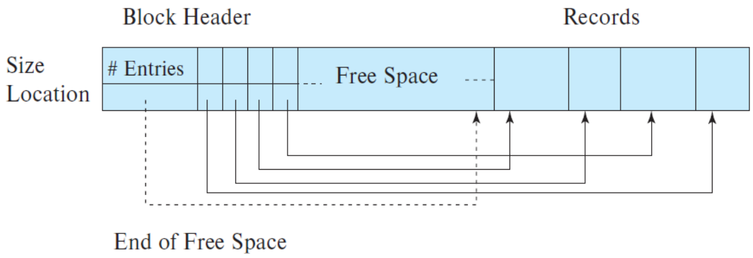

Figure 3: Slotted Page Layout

3 Advantages of Slotted Pages

- Efficient handling of variable-length records.
- Simple record insertion and deletion.
- Avoids fragmentation inside a page.
- Allows relocation of records without breaking pointers.